Part-A: CUDA Matrix Operations

Problem-1: Vector add and coalescing memory access

Q1

```
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-A$ ./vecadd00 500
Total vector size: 3840000
Time: 0.000323 (sec), GFlopsS: 11.886441, GBytesS: 142.637290
Test PASSED
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-A$ ./vecadd00 1000
Total vector size: 7680000
Time: 0.000875 (sec), GFlopsS: 8.777181, GBytesS: 105.326173
Test PASSED
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-A$ ./vecadd00 2000
Total vector size: 15360000
Time: 0.001837 (sec), GFlopsS: 8.361390, GBytesS: 100.336679
Test PASSED
```

At first glance, I thought using a loop within a thread would be inefficient. However, having each thread add ValuesPerThread elements allows each thread to do enough work to warrant its overhead, known as thread coarsening. The results show an R^2 value of .9996, indicating a strong linear correlation between the size of the vector and the execution time of the kernel. Both throughput, measured in GFlopS, and bandwidth, measured in GBytesS, decrease as the problem scales up. This indicates a problem with the implementation's memory access pattern. This program is especially sensitive to its memory management because of its low arithmetic intensity: 1 FLOP per 3 loads (4 bytes each) = 1/12 FLOPS/byte. The vecaddKernel00.cu does not use coalesced memory. For example, for ValuesPerThread=500, thread 0 works on A[0] - A[499], thread 1 works on A[500] - A[999], etc. This pattern prevents adjacent threads from loading adjacent data during the same iteration.

Q2

```
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-A$ ./vecadd01 500
Total vector size: 3840000
Time: 0.000085 (sec), GFlopsS: 45.241931, GBytesS: 542.903169
Test PASSED
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-A$ ./vecadd01 1000
Total vector size: 7680000
Time: 0.000506 (sec), GFlopsS: 15.180139, GBytesS: 182.161667
Test PASSED
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-A$ ./vecadd01 2000
Total vector size: 15360000
Time: 0.001009 (sec), GFlopsS: 15.223183, GBytesS: 182.678193
Test PASSED
```

vecaddKernel01 outperforms vecaddKernel00 at every input value and across all metrics, execution time, throughput, and bandwidth. This was achieved by using a grid-stride by each thread. This ensures for a given iteration of the kernel (not a for-loop iteration), thread 0 loads data from index 0, thread 1 from index 1, as an example for iteration 0. This coalesced access allows DRAM to deliver data in a chunk, which is likely why the best bandwidth (542 GBytesS) was so much better than with vecaddKernel00. Although GflopsS and GBytesS both worsen from ValuesPerThread 500 to 1000, they are the nearly identical from ValuesPerThread 1000 to 2000, suggesting the configuration has stabilized and will maintain a steady throughput and bandwidth even as the program scales larger. The reduction in performance from ValuesPerThread 500 to 1000 is likely due to diminishing returns of scaling up the work done by individual threads. Ideally, ValuesPerThread and the total problem size would not be tied to the same input variable, allowing ValuesPerThread to be isolated to determine the optimal value for a fixed problem size.

Kernel	ValuesPerThread	Vector Size	Time (sec)	GFlops/s	GBytes/s
vecadd00	500	3840000	0.001152	3.333222	39.998661
vecadd00	1000	7680000	0.002309	3.326338	39.916053

vecadd00	2000	15360000	0.003736	4.111328	49.335936
vecadd01	500	3840000	0.000404	9.507749	114.092992
vecadd01	1000	7680000	0.000752	10.213144	122.557722
vecadd01	2000	15360000	0.001486	10.337694	124.052329

Problem-2: Shared CUDA Matrix Multiply

Q3

```
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-A$ ./matmult00 16
Grid Dimensions: 16x16
Data dimensions: 256x256
Grid Dimensions: 16x16
Block Dimensions: 16x16
Footprint Dimensions: 16x16
Time: 0.000033 (sec), nFlops: 33554432, GFlopsS: 1012.499916
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-A$ ./matmult00 32
Grid Dimensions: 32x32
Data dimensions: 512x512
Grid Dimensions: 32x32
Block Dimensions: 16x16
Footprint Dimensions: 16x16
Time: 0.000127 (sec), nFlops: 268435456, GFlopsS: 2112.382564
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-A$ ./matmult00 64
Grid Dimensions: 64x64
Data dimensions: 1024x1024
Grid Dimensions: 64x64
Block Dimensions: 16x16
Footprint Dimensions: 16x16
Time: 0.000903 (sec), nFlops: 2147483648, GFlopsS: 2377.824513
```

Q4

```
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-A$ ./matmult01 8
Grid Dimensions: 8x8
Data dimensions: 256x256
Grid Dimensions: 8x8
Block Dimensions: 16x16
Footprint Dimensions: 32x32
Time: 0.000023 (sec), nFlops: 33554432, GFlopsS: 1450.901942
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-A$ ./matmult01 16
Grid Dimensions: 16x16
Data dimensions: 512x512
Grid Dimensions: 16x16
Block Dimensions: 16x16
Footprint Dimensions: 32x32
Time: 0.000079 (sec), nFlops: 268435456, GFlopsS: 3391.264780
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-A$ ./matmult01 32
Grid Dimensions: 32x32
Data dimensions: 1024x1024
Grid Dimensions: 32x32
Block Dimensions: 16x16
Footprint Dimensions: 32x32
Time: 0.000470 (sec), nFlops: 2147483648, GFlopsS: 4567.545261
```

Q5

Part-B CUDA Unified Memory

Q1

./q1 1

Compute time (s): 0.001953

CPU verify PASSED

(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q1 5

Compute time (s): 0.009063

CPU verify PASSED

(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q1 10

Compute time (s): 0.017489

CPU verify PASSED

(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q1 50

Compute time (s): 0.086091

CPU verify PASSED

(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q1 100

Compute time (s): 0.174514

CPU verify PASSED

```
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q1 1
Compute time (s): 0.001953
CPU verify PASSED
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q1 5
Compute time (s): 0.009063
CPU verify PASSED
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q1 10
Compute time (s): 0.017489
CPU verify PASSED
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q1 50
Compute time (s): 0.086091
CPU verify PASSED
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q1 100
Compute time (s): 0.174514
CPU verify PASSED
```

Q2

./q2 1

Scenario 1 verify PASSED

Scenario 2 verify PASSED

Scenario 3 verify PASSED

Scenario warmup times (s): 1=0.041305 2=0.000601 3=0.000032

Scenario hot times (s): 1=0.041262 2=0.000577 3=0.000013

(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q2 5

Scenario 1 verify PASSED

Scenario 2 verify PASSED

```
Scenario 3 verify PASSED

Scenario warmup times (s): 1=0.266638 2=0.007033 3=0.000202

Scenario hot times (s): 1=0.332860 2=0.007061 3=0.000233

(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q2 10

Scenario 1 verify PASSED

Scenario 2 verify PASSED

Scenario 3 verify PASSED

Scenario warmup times (s): 1=0.665087 2=0.014111 3=0.000477

Scenario hot times (s): 1=0.667568 2=0.014115 3=0.000502

(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q2 50

Scenario 1 verify PASSED

Scenario 2 verify PASSED

Scenario 3 verify PASSED

Scenario warmup times (s): 1=3.356666 2=0.070969 3=0.002525

Scenario hot times (s): 1=3.356588 2=0.070983 3=0.002566

(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q2 100

Scenario 1 verify PASSED

Scenario 2 verify PASSED

Scenario 3 verify PASSED

Scenario warmup times (s): 1=6.670520 2=0.140887 3=0.005026

Scenario hot times (s): 1=6.670588 2=0.140865 3=0.005091
```

```
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q2 1
Scenario 1 verify PASSED
Scenario 2 verify PASSED
Scenario 3 verify PASSED
Scenario warmup times (s): 1=0.041305 2=0.000601 3=0.000032
Scenario hot times (s): 1=0.041262 2=0.000577 3=0.000013
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q2 5
Scenario 1 verify PASSED
Scenario 2 verify PASSED
Scenario 3 verify PASSED
Scenario warmup times (s): 1=0.266638 2=0.007033 3=0.000202
Scenario hot times (s): 1=0.332860 2=0.007061 3=0.000233
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q2 10
Scenario 1 verify PASSED
Scenario 2 verify PASSED
Scenario 3 verify PASSED
Scenario warmup times (s): 1=0.665087 2=0.014111 3=0.000477
Scenario hot times (s): 1=0.667568 2=0.014115 3=0.000502
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q2 50
Scenario 1 verify PASSED
Scenario 2 verify PASSED
Scenario 3 verify PASSED
Scenario warmup times (s): 1=3.356666 2=0.070969 3=0.002525
Scenario hot times (s): 1=3.356588 2=0.070983 3=0.002566
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q2 100
Scenario 1 verify PASSED
Scenario 2 verify PASSED
Scenario 3 verify PASSED
Scenario warmup times (s): 1=6.670520 2=0.140887 3=0.005026
Scenario hot times (s): 1=6.670588 2=0.140865 3=0.005091
```

Q3

./q3 1

Scenario 1 verify PASSED

Scenario 2 verify PASSED

Scenario 3 verify PASSED

Scenario warmup times (s): 1=0.043749 2=0.002899 3=0.002321

Scenario hot times (s): 1=0.041136 2=0.000578 3=0.000016

(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q3 5

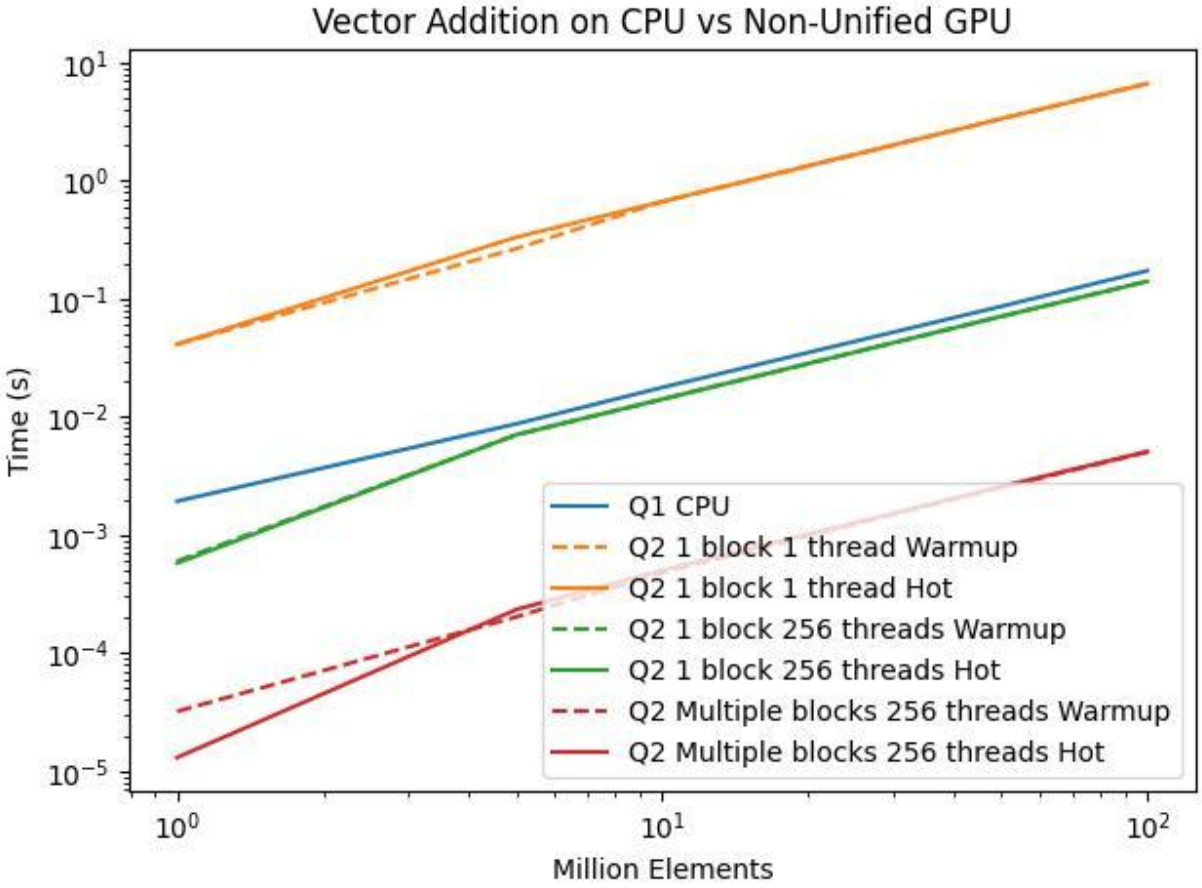
Scenario 1 verify PASSED

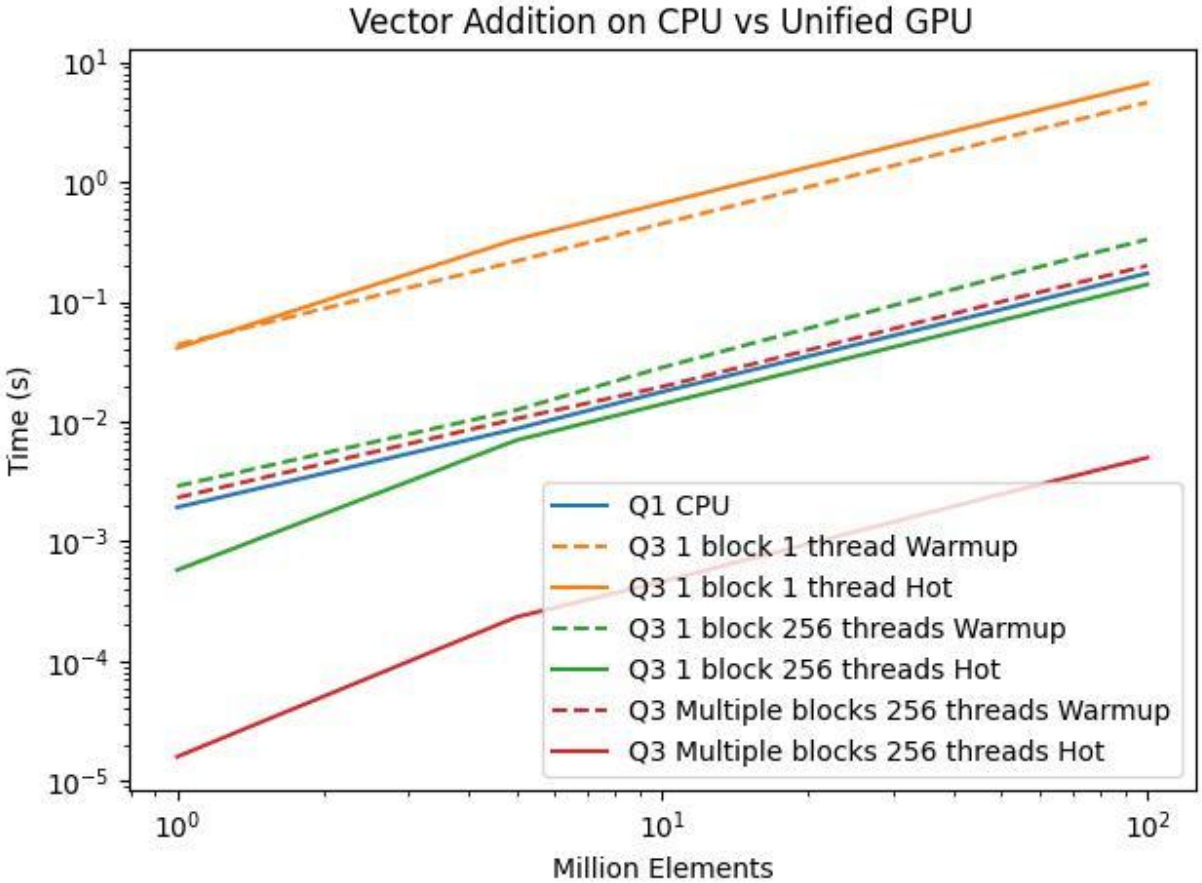
Scenario 2 verify PASSED

```
Scenario 3 verify PASSED

Scenario warmup times (s): 1=0.218390 2=0.012505 3=0.010558

Scenario hot times (s): 1=0.331891 2=0.006994 3=0.000234

(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q3 10

Scenario 1 verify PASSED

Scenario 2 verify PASSED

Scenario 3 verify PASSED

Scenario warmup times (s): 1=0.450181 2=0.028340 3=0.019570

Scenario hot times (s): 1=0.664482 2=0.014020 3=0.000463

(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q3 50

Scenario 1 verify PASSED

Scenario 2 verify PASSED

Scenario 3 verify PASSED

Scenario warmup times (s): 1=2.302808 2=0.162423 3=0.100268

Scenario hot times (s): 1=3.322097 2=0.069921 3=0.002486

(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q3 100

Scenario 1 verify PASSED

Scenario 2 verify PASSED

Scenario 3 verify PASSED

Scenario warmup times (s): 1=4.620120 2=0.331791 3=0.200413

Scenario hot times (s): 1=6.643738 2=0.139728 3=0.005009
```

```
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q3 1
Scenario 1 verify PASSED
Scenario 2 verify PASSED
Scenario 3 verify PASSED
Scenario warmup times (s): 1=0.043749 2=0.002899 3=0.002321
Scenario hot times (s): 1=0.041136 2=0.000578 3=0.000016
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q3 5
Scenario 1 verify PASSED
Scenario 2 verify PASSED
Scenario 3 verify PASSED
Scenario warmup times (s): 1=0.218390 2=0.012505 3=0.010558
Scenario hot times (s): 1=0.331891 2=0.006994 3=0.000234
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q3 10
Scenario 1 verify PASSED
Scenario 2 verify PASSED
Scenario 3 verify PASSED
Scenario warmup times (s): 1=0.450181 2=0.028340 3=0.019570
Scenario hot times (s): 1=0.664482 2=0.014020 3=0.000463
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q3 50
Scenario 1 verify PASSED
Scenario 2 verify PASSED
Scenario 3 verify PASSED
Scenario warmup times (s): 1=2.302808 2=0.162423 3=0.100268
Scenario hot times (s): 1=3.322097 2=0.069921 3=0.002486
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-B$ ./q3 100
Scenario 1 verify PASSED
Scenario 2 verify PASSED
Scenario 3 verify PASSED
Scenario warmup times (s): 1=4.620120 2=0.331791 3=0.200413
Scenario hot times (s): 1=6.643738 2=0.139728 3=0.005009
```

Q4


Vector Addition on CPU vs Non-Unified GPU
Time (s)
Million Elements
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{0}
10^{1}
10^{2}
Q1 CPU
Q2 1 block 1 thread Warmup
Q2 1 block 1 thread Hot
Q2 1 block 256 threads Warmup
Q2 1 block 256 threads Hot
Q2 Multiple blocks 256 threads Warmup
Q2 Multiple blocks 256 threads Hot



Part-A: Convolution in CUDA

Q1-Q3

```
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-C$ ./c1
Launching ConvKernel with grid (64,64,64) and block (16,16,1)
ConvKernel time: 7.954 ms
Checksum computed by GPU: 122756344698240.000000000000
Checksum OK (within tolerance)
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-C$ ./c2
Launching ConvKernel with grid (64,64,64) and block (16,16,1)
ConvKernel time: 7.986 ms
Checksum computed by GPU: 122756344698240.000000000000
Checksum OK (within tolerance)
(base) as7629@cuda-l2-instance-20260322-004309:~/CUDA1/Part-C$ ./c3
cuDNN output dimensions: N=1, C=64, H=1024, W=1024
cuDNN chose algorithm: 0
Total time of cudnnConvolutionForward: 19.096 ms
Checksum computed by GPU: 122756344698240.000000000000
Checksum OK (within tolerance)
```